

Interpret Division of Fractions

Lesson 2-1

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

12 dots

$$12 \div 3 = 4$$

In $3 \div 1/4 = 12$, the dividend is 3 — it is the total being split

Dividend

Write the definition:



$$12 \div 3 = 4$$

In $3 \div 1/4 = 12$, the divisor is $1/4$ — it is the size of each piece

Divisor

Write the definition:



$$12 \div 3 = 4$$

In $3 \div 1/4 = 12$, the quotient is 12 — there are 12 quarter-size pieces in 3

Quotient

Write the definition:

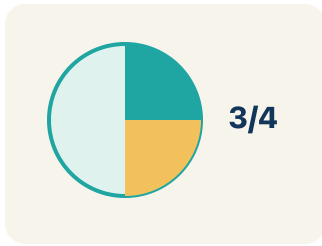
$$3/4 \rightarrow 4/3$$

flip top and bottom

The reciprocal of $1/4$ is $4/1 = 4$. Multiplying by the reciprocal gives the same result as dividing.

Reciprocal

Write the definition:



*$\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$ — each represents one equal part
of a whole*

Unit fraction

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can use models to interpret what it means to divide by a fraction.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Dividend

Divisor

Quotient

Reciprocal

Unit fraction

1 In $6 \div 1/2$, the number 6 being divided is the _____.

2 In $6 \div 1/2$, the number $1/2$ we divide by is the _____.

3 The answer to a division problem is the _____.

4 The fraction you get by flipping the numerator and denominator is the _____.

5 A fraction with a numerator of 1, like $1/5$, is a _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What does $3 \div \frac{1}{4}$ mean?

- A. How many $\frac{1}{4}$ -size pieces fit into 3
- B. 3 groups of $\frac{1}{4}$
- C. 3 minus $\frac{1}{4}$
- D. $\frac{1}{4}$ of 3

Step 1 $3 \div \frac{1}{4}$ asks: how many $\frac{1}{4}$ -size pieces fit into 3?

Step 2 The answer is 12, because each whole has 4 fourths, and $3 \times 4 = 12$.

 **Answer:** A. How many $\frac{1}{4}$ -size pieces fit into 3

 We do — together

Solve this one with your class using the same steps.

Problem: Which expression means 'how many $\frac{1}{3}$ -size pieces are in 2'?

- A. $2 \div \frac{1}{3}$
- B. $2 \times \frac{1}{3}$
- C. $\frac{1}{3} \div 2$
- D. $2 + \frac{1}{3}$

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: A rope is 4 feet long. How many $\frac{1}{2}$ -foot pieces can be cut from it?

- A. 8 pieces
- B. 2 pieces
- C. 4 pieces
- D. $\frac{1}{2}$ piece

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Agent Cole has 8 feet of evidence rope and cuts it into $\frac{1}{3}$ -foot pieces for tagging. What does $8 \div \frac{1}{3}$ mean?

- A. How many $\frac{1}{3}$ -foot pieces fit into 8 feet
- B. $\frac{1}{3}$ of the 8 feet of rope
- C. 8 feet shortened by $\frac{1}{3}$ of a foot
- D. 8 pieces that are each $\frac{1}{3}$ foot long

Show your work:

2. Picture clue: a 5-inch fingerprint strip is split into $\frac{1}{6}$ -inch slivers under the microscope. Which division does the picture show, and what is the count?

- A. $5 \div \frac{1}{6} = 30$ slivers
- B. $5 \div 6 = \frac{5}{6}$ of a sliver
- C. $\frac{1}{6} \div 5 = \frac{1}{30}$ sliver
- D. $5 \times \frac{1}{6} = \frac{5}{6}$ of an inch

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Dividing 6 by $\frac{1}{2}$ gives 12, but dividing 6 by 2 gives 3. How can dividing by a smaller number ($\frac{1}{2}$) give a bigger answer than dividing by a larger number (2)? Explain using a real-world example.

Sentence starter: When I divide 6 by 2, I am _____. When I divide 6 by $\frac{1}{2}$, I am _____. The answer is bigger because _____.

Show your work:

Reflect — Exit Ticket

What does $5 \div \frac{1}{4}$ mean, and what is the quotient?

- A. How many $\frac{1}{4}$ -size pieces fit into 5; quotient is 20
- B. 5 groups of $\frac{1}{4}$; quotient is $\frac{5}{4}$
- C. $\frac{1}{4}$ of 5; quotient is $\frac{5}{4}$
- D. 5 minus $\frac{1}{4}$; quotient is $4\frac{3}{4}$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Dividend
2. **Notes 2:** Divisor
3. **Notes 3:** Quotient
4. **Notes 4:** Reciprocal
5. **Notes 5:** Unit fraction
6. **We Try (notes):** A. $2 \div 1/3$ — *Finding how many $1/3$ -size pieces fit into 2 is represented by $2 \div 1/3$. The answer is 6.*
7. **You Try (notes):** A. 8 pieces — $4 \div 1/2 = 4 \times 2 = 8$ pieces. *Each whole foot contains 2 halves, and $4 \times 2 = 8$.*
8. **Try It 1:** A. How many $1/3$ -foot pieces fit into 8 feet — *Dividing by $1/3$ asks how many $1/3$ -foot pieces fit into 8 feet. Each foot holds 3 thirds, so $8 \times 3 = 24$ pieces.*
9. **Try It 2:** A. $5 \div 1/6 = 30$ slivers — *How many $1/6$ -inch slivers fit into 5 inches? Each inch holds 6 sixths, so $5 \times 6 = 30$ slivers.*
10. **Exit Ticket:** A. How many $1/4$ -size pieces fit into 5; quotient is 20 — $5 \div 1/4$ means 'how many $1/4$ -size pieces fit into 5.' *Since each whole has 4 fourths, $5 \times 4 = 20$.*